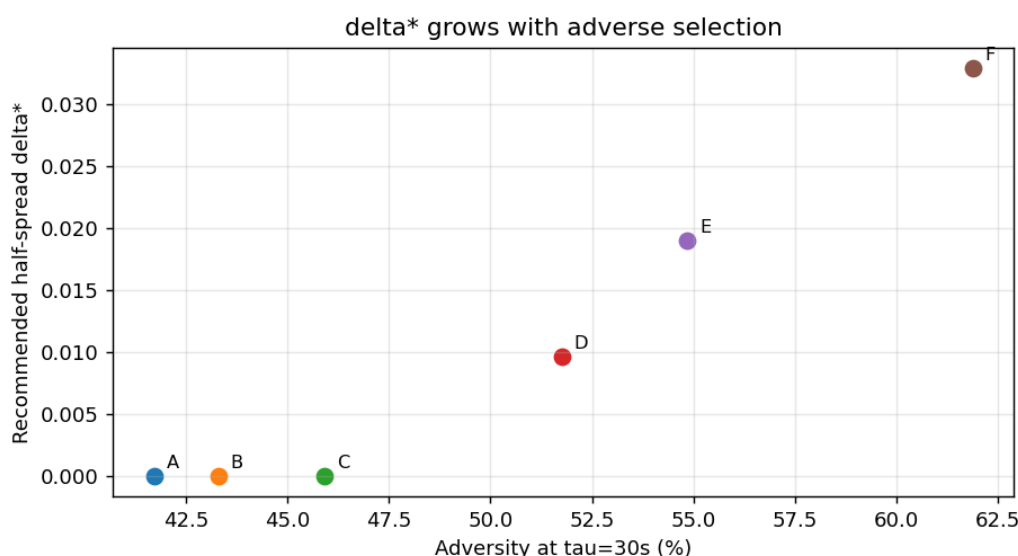


Task 2 - Client Profitability and Spread Recommendation

Expected PnL: Per horizon we report the sample mean of $\text{side} \cdot V \cdot (M_t - T_p)$ (eq. 5); agg_pnl is the sample mean of $\text{side} \cdot V \cdot (\bar{M} - T_p)$ with uniform $1/6$ weights (eq. 6), where $\bar{M}$ is the mean of $M_5 \dots M_{30}$. 'Expected' = sample average per trade.

Results table

client	$\tau=5$	$\tau=10$	$\tau=15$	$\tau=20$	$\tau=25$	$\tau=30$	agg_pnl	δ^*	class
A	1.268	1.540	1.737	2.061	2.293	2.485	1.897	0.00000	profitable
B	1.211	1.342	1.542	1.726	1.909	2.059	1.632	0.00000	profitable
C	1.077	1.165	1.190	1.162	1.229	1.216	1.173	0.00000	profitable
D	0.868	0.682	0.493	0.177	-0.076	-0.347	0.299	0.00961	profitable
E	0.653	0.303	-0.108	-0.572	-1.069	-1.614	-0.401	0.01898	costly
F	0.445	-0.176	-0.917	-1.751	-2.614	-3.629	-1.440	0.03287	costly



Classification (net-profitable iff $\text{agg_pnl} \geq 0$)

Profitable: A (1.897), B (1.632), C (1.173), D (0.299). Costly: E (-0.401), F (-1.440). A, B, C earn positive PnL at every horizon. D is positive in aggregate but its per-horizon PnL turns negative by $\tau=25$ -30s, warning that its edge decays. E and F lose money in aggregate: their PnL flips negative early (E by $\sim 15s$, F by $\sim 10s$) and deteriorates sharply - classic toxic flow.

Minimum half-spread δ^*

If we quote $M_0 \pm \delta$, the effective fill price is $T_p^{\text{eff}} = M_0 - \text{side} \cdot \delta$, so eq. 6 becomes $E[\text{AggPnL}(\delta)] = A + \delta \cdot E[V]$, with $A = E[\text{side} \cdot V \cdot (\bar{M} - M_0)]$ the aggregate PnL at the mid ($\delta=0$). Requiring non-negativity gives $\delta^* = \max(0, -A / E[V])$. Results: A=B=C=0 (already profitable at the mid), D=0.0096, E=0.0190, F=0.0329.

Link to Task 1 adversity

δ^* is precisely the half-spread that offsets adverse selection. The more adverse a client (Task 1), the more the mid drifts against us after a fill, the more negative A becomes, and the larger δ^* must be. The ranking is identical: A,B,C (benign) need no extra spread; D, E, F (rising adversity) require monotonically larger δ^* . δ^* is the monetary translation of the adversity curve.